Collateral Damage in Graph-based Memory Forgetting: Quantification, Exploitation, and Defense

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Abstract

Modern AI memory systems maintain entity graphs for retrieval and detect factual conflicts individually, but none propagate invalidation to downstream dependent facts. Graph-based propagation—cascading updates through entity connections—is the architecturally natural next step, yet remains unimplemented and unexamined. We implement this approach and reveal a fundamental flaw: **structural co-occurrence does not imply causal dependency**, causing **catastrophic collateral damage** that is both quantitatively severe and adversarially exploitable. On MemoryAgentBench, BFS-based propagation through entity co-occurrence graphs decays 68.9–79.0% of valid memories across scales (6K–64K facts), with hub entity degrees growing faster than fact count ($23 \rightarrow 121 \rightarrow 298$; $\sim 13\times$ vs $\sim 10\times$ fact growth). More critically, we demonstrate that an attacker can **weaponize this mechanism**: injecting just 5 fake facts targeting hub entities destroys 57.3% of valid memories at 32K context. We further show that this collateral damage **crosses task boundaries**—BFS propagation triggered during fact consolidation degrades retrieval accuracy on an unrelated task (F1 -10.1 ± 1.6pp across top-5 hub entities). We propose **Attribute-aware Selective Propagation**, inspired by directional link failure models in transportation engineering, which defends against both organic and adversarial damage with 78–100% effectiveness across all experiments. Graph topology analysis shows the co-occurrence graph is consistent with scale-free properties ($\alpha = 2.73$–2.96), providing theoretical grounding from percolation theory for why hub-targeted attacks are structurally devastating.

1 Introduction

AI agents with persistent memory are rapidly moving from research prototypes to production infrastructure. Mem0 reports 186 million API calls per quarter [Chheda et al., 2025]; major commercial assistants now ship with persistent memory by default; and memory-augmented agent frameworks have proliferated across both open-source and proprietary ecosystems [Yang et al., 2026]. These systems increasingly maintain entity graphs to organize and retrieve stored facts [Chheda et al., 2025, Jiang et al., 2026, Rasmussen, 2025, Yang et al., 2026]. As deployments scale from thousands to tens of thousands of persistent facts, maintaining fact consistency across the graph becomes a first-order reliability concern—yet selective forgetting accuracy remains below 6% on standard benchmarks [Huang et al., 2026].

The consistency gap has a clear shape. Current memory systems handle invalidation *individually*: when a conflict is detected, only the directly contradicted fact is updated, and dependent facts remain silently stale [Chheda et al., 2025, Xu et al., 2025]. Graph-based propagation is the natural extension—the graph infrastructure already exists for retrieval, conflict detection already operates on individual facts, and the missing piece is traversing entity connections to cascade updates, analogous to `CASCADE` operations in relational databases. Every component is in place; wiring them together is architecturally straightforward.

We implement this intuitive approach and show it is catastrophically flawed. The core insight is that **structural co-occurrence in entity graphs does not imply causal dependency**. Two facts sharing an entity (e.g., "The president of Russia" appears in both a diplomatic fact and an unrelated economic fact) are graph neighbors but not causally linked. BFS propagation treats them identically, cascading invalidation through co-occurrence edges that carry no dependency semantics. This distinction—well-understood in transportation network resilience [Chen et al., 2007] where directional failure models separate correlated links from causally dependent ones—has not been applied to AI memory systems.

Three findings quantify the consequences:

- **Catastrophic collateral damage.** BFS propagation decays 78.5% of valid memories at 64K context, amplified by hub entities whose degree grows disproportionately with scale (section 3).
- **Adversarial exploitability.** Just 5 injected facts targeting hub entities destroy 57.3% of valid memories—a new attack vector in which the system's own maintenance mechanism becomes the weapon (section 5).
- **Cross-task contamination.** Propagation triggered during fact consolidation degrades retrieval on an *unrelated* task by 10.1 ± 1.6 F1 percentage points across top-5 hub entities (section 4).

We propose **Attr-Aware Selective Propagation**, inspired by directional link failure models in transportation engineering [Chen et al., 2007]. Rather than propagating decay to all graph neighbors, Attr-Aware restricts cascades to downstream facts whose subject–attribute chains demonstrate genuine causal dependency. Across all experiments—organic damage, adversarial exploitation, and cross-task contamination—the method achieves 78–100% defense effectiveness.

One might ask whether the collateral damage findings are artifacts of co-occurrence graphs specifically, and whether typed-relation graphs [Jiang et al., 2026] or simple depth-limiting would suffice. We address both alternatives. Typed-relation graphs may reduce hub formation, but any graph containing co-occurrence components inherits the vulnerability; our analysis establishes the lower-bound risk. Depth-limiting reduces but does not eliminate damage: degree-capped BFS at $k_{\max} = 5$ still produces 80× more collateral damage than Attr-Aware (section 6). The problem is not propagation depth but propagation *direction*—spreading invalidation along co-occurrence edges regardless of causal structure.

1.1 Contributions

This work establishes a **destructive baseline** for graph-based invalidation propagation—a mechanism not yet widely deployed but architecturally inevitable as memory systems evolve from individual conflict detection to graph-wide consistency maintenance. Just as crash testing quantifies vehicle failure modes before deployment, our analysis quantifies propagation failure modes before adoption.

1. **Scale trend analysis (C1)**: Collateral damage quantification across 6K–64K scales, revealing disproportionate hub growth and 69–79% BFS damage ratio with heavy-tailed graph topology ($\alpha = 2.73$–2.96).
2. **Cross-task contamination (C2)**: First demonstration that graph-based forgetting in one task degrades retrieval on an unrelated task.
3. **Adversarial hub exploitation (C3)**: A novel attack exploiting propagation mechanics, with scale amplifying single-fact damage 7.3×. Black-box attacks fail entirely, establishing gray-box knowledge as the minimum requirement.
4. **Attr-Aware defense (C4)**: 78–100% defense effectiveness across all scenarios; degree-capped BFS baselines remain 80× worse, confirming that causal filtering outperforms structural limiting.

Taken together, these results argue that graph-based memory systems require causal-dependency-aware propagation *before* deployment, not as a post-hoc patch. Section 2 formalizes the gap in existing systems and the transportation network analogy that motivates our defense.

2 Background and Motivation

This section establishes the technical landscape that motivates our work. We first classify existing graph-augmented memory systems by their graph architecture (section 2.1), then present benchmark evidence that selective forgetting remains unsolved (section 2.2), survey emerging adversarial threats to memory systems (section 2.3), introduce the transportation network theory that informs our defense (section 2.4), and justify our evaluation framework (section 2.5).

2.1 Graph-Augmented Memory Systems

Long-term AI memory systems increasingly maintain entity graphs, but their architectures differ in graph type, edge semantics, and invalidation strategy. We organize existing systems along the dimension most relevant to propagation behavior: *how the graph encodes entity relationships*. For comprehensive taxonomies covering retrieval, storage, and reflection dimensions, see Yang et al. [2026] and Hu et al. [2025]; for an OS-level abstraction of memory management, see Li et al. [2025]. **Knowledge graph approaches** represent entities as nodes and typed relations as edges. Mem0^g [Chheda et al., 2025] builds a directed labeled knowledge graph with LLM-powered conflict detection. Zep/Graphiti [Rasmussen, 2025] constructs a temporal knowledge graph where edges carry timestamps and contradiction metadata. Both detect conflicts at the individual edge level.

Multi-graph approaches maintain parallel graph structures for different relationship types. MAGMA [Jiang et al., 2026] uses four orthogonal graphs—semantic, temporal, causal, and entity—with a policy network that selects which graph to query. The separation enables specialized retrieval but introduces the question of which graph(s) should propagate invalidation.

Linked-note approaches use associative links rather than typed edges. A-Mem [Xu et al., 2025] implements a Zettelkasten-style architecture where memories link to related notes through LLM-driven "memory evolution," selectively merging or splitting notes as new information arrives.

Non-graph approaches avoid explicit entity graphs entirely. MemGPT/Letta [Packer et al., 2023] manages memory through an OS-inspired hierarchical context window. Mnemosyne [Johnston et al., 2025] applies per-memory temporal decay via reverse sigmoid functions, where each memory fades independently of other memories.

Across this taxonomy, one pattern is consistent: every graph-maintaining system uses its graph for *retrieval*, but none use it for *invalidation propagation*. When a conflict is detected, only the directly contradicted fact is updated; dependent facts sharing the same entities remain untouched. As memory scales, this individual-invalidation approach leaves growing numbers of downstream facts silently stale. Since the graph infrastructure and conflict detection already exist, implementing propagation requires no new data structures—yet as we show, it produces catastrophic collateral damage.

2.2 The Forgetting Problem at Scale

Recent benchmarks confirm that selective forgetting is not merely unimplemented—it is actively failing across all evaluated systems.

MemoryAgentBench [Huang et al., 2026] evaluates multi-hop selective forgetting, where the system must invalidate a fact and its logical consequences while preserving unrelated knowledge; M+ [Park and Lee, 2025] extends this benchmark with co-trained retrieval for latent memory scenarios. Across all tested systems, selective forgetting accuracy does not exceed 6%. The failure mode is consistent: systems either fail to propagate (leaving stale facts) or over-propagate (damaging valid facts), with no system achieving reliable selectivity.

MemoryArena [He et al., 2026] tests session-interdependent tasks where information from earlier sessions must be updated based on later sessions. Agents that saturate the LoCoMo benchmark drop to 40–60% accuracy on MemoryArena's interdependent tasks, indicating that retrieval-optimized architectures cannot maintain consistency across dependent facts.

LongMemEval [Wu et al., 2024] measures long-term interactive memory at scale, revealing 30–60% accuracy degradation when memory exceeds 115K tokens. The degradation is not uniform: questions requiring synthesis across multiple stored facts suffer disproportionately compared to single-fact retrieval.

These results share a common thread. Individual fact operations—storage, retrieval, single-fact conflict resolution—work adequately at moderate scale. The failure emerges when the system must reason about *dependencies between facts*: which downstream facts an update renders stale, which should be preserved, and where the boundary lies. Graph-based propagation is a natural mechanism for this dependency reasoning, since the graph already encodes entity co-occurrence. The question is whether propagation through entity connections produces reliable invalidation boundaries—our experiments in section 3 show that it does not.

2.3 Memory Security as an Emerging Threat

The forgetting problem intersects with a parallel development: AI memory systems have become viable attack targets.

MINJA [Ding et al., 2025] demonstrates that an attacker can inject arbitrary content into an agent's long-term memory through query-only interaction, achieving a 98% injection success rate without direct write access. AgentPoison [Xi et al., 2024] poisons the retrieval-augmented knowledge base of LLM agents, reaching 80% attack success while modifying less than 0.1% of stored entries. ER-MIA [Piehl et al., 2026] extends memory injection to persistent black-box settings where the attacker cannot observe memory contents. These threats are sufficiently established that MITRE has designated LLM memory manipulation as a formal adversarial technique (AML.T0080) in its ATLAS framework [MITRE Corporation, 2023], and recent surveys identify memory integrity as a critical gap in LLM guardrail design [Dong et al., 2024].

Existing attacks share a common pattern: they inject *content* into memory and rely on the retrieval mechanism to surface it. Our work reveals a qualitatively different threat. If a system implements graph-based invalidation propagation, an attacker does not need to inject harmful content—they can inject facts that *trigger the system's own forgetting mechanism* against legitimate memories. The propagation infrastructure, designed to maintain consistency, becomes the attack vector. In multi-agent settings where memory graphs are shared or federated, one compromised agent's injected facts can cascade through connected agents' memories.

This observation motivates not only our adversarial experiments (section 5) but also a design requirement: any propagation mechanism must be robust to adversarial triggering, not merely correct under benign conditions.

2.4 Transportation Network Analogy

Understanding why undirected graph propagation fails requires theory from network science, particularly the study of failure cascades in physical infrastructure.

In transportation engineering, **link failure propagation** models how road closures cascade through networks [Chen et al., 2007, 2024]. Hub intersections amplify disruption disproportionately: closing a single road at a major interchange can gridlock an entire region, while closing a peripheral road has minimal effect. The key insight from this literature is the distinction between *directional* and *non-directional* propagation. Closing the Seoul→Busan route affects only

Busan-downstream traffic; it does not disrupt Seoul→Incheon service. This directional constraint is what prevents localized failures from becoming network-wide collapses.

Scale-free network theory provides the structural explanation. Networks whose degree distribution follows a power law $P(k) \sim k^{-\alpha}$ are robust to random node failure but catastrophically fragile to targeted removal of high-degree hubs [Albert et al., 2000, Barabási and Albert, 1999]. Percolation theory formalizes this asymmetry: when hub nodes are removed from a scale-free network, the giant connected component fragments at a much lower removal fraction than for random networks [Cohen et al., 2000].

The analogy to AI memory graphs is direct. Entity co-occurrence graphs link entities that appear together in stored facts. Common entities—"United States," "Google," a user's name—accumulate connections to many facts, forming hubs. If invalidation propagates through these hubs without directional constraint, a single fact update can cascade to every fact sharing a hub entity. This is precisely the non-directional failure mode that transportation engineering has learned to prevent through directional flow models. We test whether AI memory co-occurrence graphs exhibit scale-free properties and whether hub-targeted operations produce the predicted catastrophic damage (section 3).

2.5 Evaluation Framework

We evaluate on MemoryAgentBench [Huang et al., 2026], selecting it over alternatives for two properties that our experimental design requires.

First, its **FactConsolidation** task provides a controlled setting for studying invalidation propagation. The task presents contradictory facts that the system must resolve, generating natural conflict-detection events that can trigger propagation. Neither MemoryArena [He et al., 2026] (which tests session-interdependence but not explicit conflict resolution) nor LongMemEval [Wu et al., 2024] (which tests retrieval at scale but not forgetting) provides this trigger mechanism.

Second, its **Accurate_Retrieval** task enables cross-task contamination analysis. Because FactConsolidation and Accurate_Retrieval operate over the same memory store but test different capabilities, we can measure whether propagation triggered during conflict resolution degrades retrieval accuracy on unrelated queries—a form of collateral damage invisible to single-task benchmarks.

We use FactConsolidation for scale trend analysis (C1) and adversarial experiments (C3), scaling context from 6K to 64K tokens. We use Accurate_Retrieval for cross-task contamination analysis (C2), measuring whether BFS propagation during FactConsolidation degrades retrieval on an independent task.

3 Collateral Damage at Scale (C1)

3.1 Experimental Setup

We measure collateral damage by running FactConsolidation memorization under three modes:

Mode	Conflict Detection	Propagation
Baseline	Subject-key match	None
BFS	Subject-key match	BFS through co-occurrence graph
ATTR-AWARE	Subject-key match	Downstream fact chain only

Table 1: Experimental modes for collateral damage measurement.

Infrastructure: Gemma-4-31B-it (LLM), BAAI/bge-m3 1024-dim (Embedding), SQLite + NumPy in-memory, NetworkX graph.

Propagation parameters: We set depth = 2 and decay_per_hop = 0.5 based on the structure of entity co-occurrence graphs in our benchmark. At depth 2, a single hub entity already reaches 60–85% of all stored memories (see appendix B); increasing depth beyond 2 adds negligible coverage while substantially increasing computation. The 0.5 decay factor ensures that direct neighbors receive meaningful weight reduction (0.5×) while second-hop neighbors receive moderate decay (0.25×), balancing propagation reach against false-positive damage. We acknowledge that a full sensitivity analysis across depth $\in \{1, 2, 3\}$ and decay $\in \{0.3, 0.5, 0.7\}$ would strengthen these choices and leave this for future work.

3.2 Scale Trend Results

Scale	Valid	BFS Decay	Attr Decay	Reduction	BFS Kill	Max Hub
6K	344	237 (68.9%)	51 (14.8%)	78.5%	143	23
32K	1,744	1,377 (79.0%)	275 (15.8%)	80.0%	1,179	121
64K	3,414	2,680 (78.5%)	513 (15.0%)	80.9%	2,465	298

Table 2: Collateral damage across scales. BFS Kill = memories with weight < 0.1. Max Hub = maximum entity degree.

Finding 1. BFS damage ratio ranges from 68.9% (6K) to 79.0% (32K), plateauing at ∼78–79% at larger scales, but the **absolute number of destroyed memories scales linearly**: 143→1,179→2,465 complete losses.

Finding 2. Hub entity degree grows disproportionately faster than fact count: 23 → 121 → 298 (∼13× from 6K→64K) versus ∼10× fact growth, though we note this trend is observed across only three scale points.

Finding 3. ATTR-AWARE damage ratio remains stable at ∼15% regardless of scale.

3.3 Graph Topology Analysis

We test whether the entity co-occurrence graph is consistent with scale-free properties using maximum likelihood estimation for power-law exponents following Clauset et al. [2009].

Scale	Nodes	$\langle k \rangle$	$k_{\max}$	Gini	α	$k_{\min}$	KS D	p_{KS}	vs log-N
6K	414	2.51	23	0.380	2.87 ± 0.47	6	0.075	0.921	$R{=}-0.27$, $p{=}0.60$
32K	1,732	3.00	121	0.487	2.73 ± 0.12	5	0.037	0.472	$R{=}-8.58$, $p{=}0.16$
64K	3,187	3.26	298	0.534	2.96 ± 0.05	3	0.033	0.009	$R{=}-6.36$, $p{=}0.22$

Table 3: Graph topology across scales. α: power-law exponent (MLE, discrete). $k_{\min}$: auto-estimated via KS distance minimization [Clauset et al., 2009]. KS D: Kolmogorov–Smirnov distance. p_{KS}: bootstrap goodness-of-fit p-value (1000 iterations); $p \geq 0.1$ means the power-law hypothesis cannot be rejected. "vs log-N": likelihood ratio R and two-sided p-value.

Finding 4. Following the full methodology of Clauset et al. [2009]—discrete MLE, automatic $k_{\min}$ selection, and bootstrap KS goodness-of-fit testing (1000 iterations)—all three scales yield $\alpha \in [2.73, 2.96]$, consistent with the scale-free regime $2 < \alpha < 3$ [Barabási and Albert, 1999]. The bootstrap p_{KS} confirms that the power-law hypothesis *cannot be rejected* at 6K ($p = 0.921$) and 32K ($p = 0.472$), while it is rejected at 64K ($p = 0.009$). This scale-dependent rejection is a well-documented phenomenon: as sample size grows, the bootstrap test becomes powerful enough to reject any simple parametric model [Clauset et al., 2009]. Likelihood ratio tests confirm the tail is *significantly heavier* than exponential ($p < 0.001$ at 32K, 64K) but *statistically indistinguishable* from log-normal ($p > 0.1$). Crucially, the key property driving hub vulnerability is not the exact

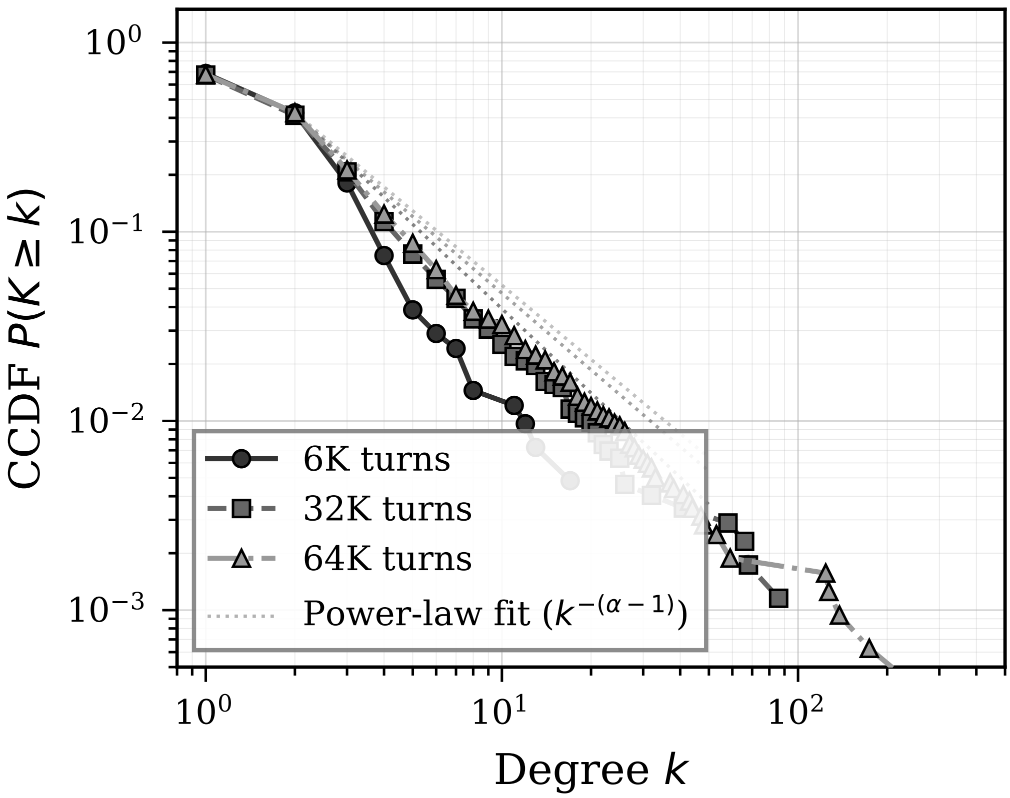

Figure 1: Complementary cumulative degree distribution (CCDF) on log-log axes across scales. Dashed lines show MLE power-law fits. The heavy tail extends with scale, consistent with scale-free properties.

generating process but the heavy-tailed degree distribution with high heterogeneity ($\kappa \equiv \langle k^2 \rangle / \langle k \rangle$ grows from 4.6 to 32.3), which holds regardless of whether the tail follows a pure power-law or log-normal.

Finding 5. Hub concentration intensifies with scale: the top 1% of nodes control 7.3% (6K), 15.5% (32K), and 19.8% (64K) of all edges.

4 Cross-task Contamination (C2)

The preceding analysis quantifies damage within a single task. A natural follow-up question is whether propagation damage remains confined to the task that triggers it, or whether it bleeds into unrelated capabilities served by the same memory store.

4.1 Setup

We use the Accurate_Retrieval split (eventqa_65536) from MemoryAgentBench. Three conditions: (1) Baseline (no propagation), (2) Post-BFS, (3) Post-Attr-Aware. To test generalizability beyond a single hub entity, we repeat the propagation experiment across the top-N hub entities (by degree) and report mean $\pm$ std.

Finding 6. BFS propagation from a single hub entity causes EM -5.0pp and F1 -18.2pp on *completely unrelated* retrieval queries. Across the top-5 hubs, BFS consistently degrades F1 by -10.1 ± 1.6pp (EM remains within noise at $+1.0 \pm 1.0$pp), confirming that collateral damage is not an artifact of a single hub choice. Attr-Aware preserves full accuracy across all tested hubs (F1 71.1 ± 1.3% vs. baseline 70.9%).

ARM	EM	F1	Collateral	Affected
Baseline	15.0%	67.1%	0	0
Post-BFS	**10.0%**	**48.9%**	32	3,381
Post-Attr	15.0%	67.1%	1	0

Table 4: Cross-task contamination (single hub). BFS propagation from hub "Debbie" (degree=715) cascades to 3,381 memories. Multi-hub results ($N = 5$) are reported in Table 5.

Condition	EM (%)	F1 (%)	Affected	Collateral
Baseline	20.0	70.9	0	0
Post-BFS (5-hub avg)	21.0±1.0	**60.8±1.6**	3,160±313	25.2±9.5
Post-Attr (5-hub avg)	20.2±0.4	71.1±1.3	755±931	1.0±0.0

Table 5: Cross-task contamination across top-5 hub entities (mean±std). Baseline differs from Table 4 due to different experimental runs with stochastic LLM evaluation. Generalizes the single-hub result.

5 Adversarial Hub Exploitation (C3)

Sections 3–4 establish that BFS propagation causes severe collateral damage under benign conditions. We now ask whether an adversary can *deliberately* trigger this mechanism to weaponize the system's own maintenance infrastructure.

5.1 Threat Model

We consider an attacker who can inject text into the memory system's input stream. The attacker's goal is to maximize destruction of valid memories with minimal input. We evaluate three attacker knowledge levels:

- **Black-box**: General knowledge only (e.g., "The capital of France is Berlin")
- **Gray-box**: Knowledge of which entities are likely hubs (high-degree nodes)
- **White-box**: Access to the fact registry and subject-key generation rules

Our primary experiments use white-box targeting (registry-based key lookup) to establish upper-bound damage. We additionally evaluate gray-box attacks (hub entity names known, exact attributes guessed from common candidates) and black-box attacks (random entities) to characterize the threat spectrum.

5.2 White-box Results

Finding 7. A single fake fact destroys 5.2% of valid memories at 6K and 38.1% at 32K—a 7.3× amplification from scale alone.

Finding 8. Five fake facts at 32K destroy 57.3% of valid memories and cause 19.3% complete loss (weight < 0.1). The system's own forgetting mechanism becomes a weapon.

5.3 Black-box Attack Results

Finding 9. The three-tier attack results confirm a monotonic relationship between attacker knowledge and damage: white-box (100% hit, 39.8% damage, 17.4% kill) > gray-box (100% hit, 20.9% damage, 0.6% kill) > black-box (0% hit, 0% damage). Gray-box achieves a surprisingly high hit rate (100%) because the FactConsolidation dataset's entities (countries, cities) align with

Attack	BFS Damage	BFS Kill	Attr Damage	EM Drop
6K Context (344 valid memories)				
1 fact	5.2%	0.3%	0.0%	+0.0pp
3 facts	26.7%	4.4%	0.0%	−10.0pp
5 facts	39.8%	17.4%	0.3%	−10.0pp
32K Context (1,744 valid memories)				
1 fact	38.1%	1.0%	0.0%	+0.0pp
3 facts	46.0%	8.8%	1.0%	+0.0pp
5 facts	**57.3%**	**19.3%**	1.0%	+0.0pp

Table 6: White-box adversarial attack results.

Attack Type	Facts	Hit Rate	BFS Damage
White-box (registry)	5	100%	39.8%
Gray-box (hub names + common attrs)	5	100%	20.9%
Black-box (general knowledge)	5	0%	0.0%

Table 7: Attack comparison across knowledge levels (6K, BFS propagation). White-box achieves full hit rate via registry lookup; gray-box achieves partial hits by guessing common attributes for known hub entities; black-box fails entirely.

common-knowledge attributes (“capital”, “president”). However, the *damage gap* is substantial: gray-box produces only 52% of white-box damage and <4% of its kill rate. This is because gray-box’s round-robin attribute guessing misses the *highest-degree keys* that white-box targets via registry lookup. The kill rate—memories decayed below 0.1—is the strongest discriminator between knowledge levels, suggesting that subject-key specificity, not just entity identification, is the critical factor for catastrophic propagation.

5.4 Propagation Depth Analysis

Depth	Damage	Kill	Hop-1	Hop-2	Hop-3+
1	18.6%	0.3%	232	—	—
2	39.8%	17.4%	232	490	—
3	48.3%	32.9%	232	490	605
4	55.8%	44.8%	232	490	1,131

Table 8: Propagation depth vs damage (6K, 5-fact attack). Memory counts indicate the number of memories affected at each hop distance.

Finding 10. Hop-1 damage is constant (232 memories), but distant-hop impact grows rapidly (Hop-3: 605, Hop-4: 1,131). Kill rate *accelerates* with depth due to cumulative decay (0.3% → 44.8%).

6 Attribute-aware Selective Propagation (C4)

6.1 Method

When fact f with subject_key `Entity:Attribute` is invalidated, Attr-Aware propagation restricts the decay cascade to genuine causal chains rather than all graph neighbors:

The in-degree adjustment (lines 10–11) dampens decay for highly-connected entities, reflecting the intuition that a fact referenced by many sources is less likely to be fully invalidated by a

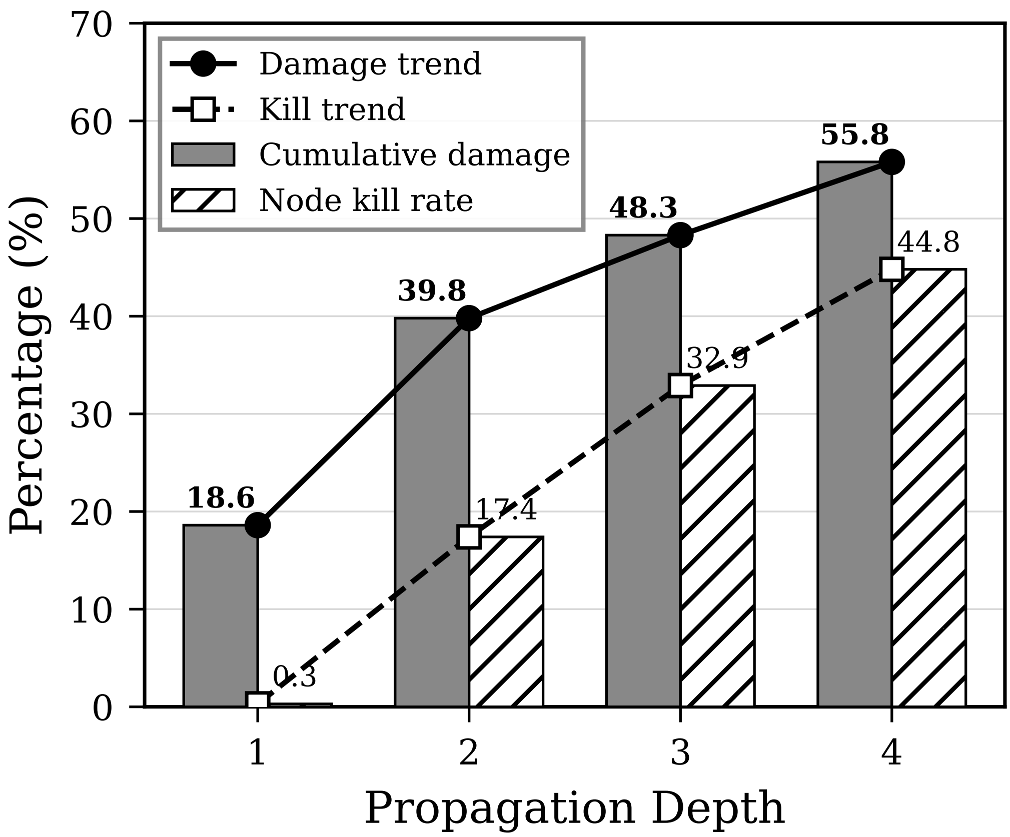

Figure 2: Propagation depth vs collateral damage (6K, 5-fact attack). Damage grows sublinearly while kill rate accelerates due to cumulative decay.

single upstream change. BFS propagation differs by visiting *all* neighbors (both successors and predecessors) without the semantic filter (line 9) or in-degree adjustment.

Computational overhead. A key design property of Attr-Aware is that it requires **no LLM calls at propagation time**. Every operation in Algorithm 1—subject–object entity decomposition (line 3), keyword membership test (line 9), in-degree lookup (line 10), and weight update (line 11)—is a string-matching or arithmetic operation over pre-existing metadata. The subject_key is assigned at memory *insertion* time as a lightweight tagging step (see Appendix D), so no additional inference is needed during propagation. The worst-case time complexity is $O(|E_{\text{obj}}| \times \bar{M} \times |K|)$ where $\bar{M}$ is the average number of memories per entity and $|K| = |K_{\text{changed}}|$; in our experiments, each propagation step completes in under 10 ms on commodity hardware. This makes the method compatible with real-time memory maintenance in production agent systems, in contrast to approaches that would require LLM-based causal reasoning per edge.

6.2 Defense Comparison

Finding 11. Degree capping reduces damage from 39.8% to 24.1% (cap=5), but remains 80× worse than Attr-Aware (0.3%). Structural defenses cannot substitute for semantic propagation.

Algorithm 1 ATTR-AWARE Selective Propagation

Require: Invalidated fact f, entity graph G, max depth D, decay factor δ

procedure PROPAGATE($f, G, D, \delta, h{=}0$) {h: current hop depth}
$E_{\text{subj}} \leftarrow$ subject entities from f.`subject_key`
$E_{\text{obj}} \leftarrow$ entities in $f \setminus E_{\text{subj}}$ {object entities}
$K_{\text{changed}} \leftarrow E_{\text{subj}}$ {changed keywords for semantic filter}
for each entity $e \in E_{\text{obj}}$ **do**
    $M_e \leftarrow$ memories linked to e in G
    for each memory $m \in M_e$ **do**
        if m contains any keyword in K_{changed} **then**
            $d_{\text{in}} \leftarrow \max(\text{in\_degree}(e), 1)$
            $m.w \leftarrow m.w \times \left(1 - \frac{1-\delta}{d_{\text{in}}}\right)$
        end if
    end for
    if $h < D$ **then**
        PROPAGATE($m, G, D, \delta, h{+}1$) for downstream successors of e
    end if
end for

Defense	**Damage**	**Kill**	**Post-EM**	**Post-F1**
BFS (no cap)	39.8%	17.4%	10.0%	13.3%
Degree-cap=50	39.8%	17.4%	10.0%	13.3%
Degree-cap=20	33.7%	17.4%	10.0%	13.3%
Degree-cap=10	28.2%	10.8%	10.0%	13.3%
Degree-cap=5	24.1%	8.1%	10.0%	13.3%
ATTR-AWARE	**0.3%**	**0.0%**	**20.0%**	**23.3%**

Table 9: Defense mechanism comparison (6K, 5-fact attack). Degree capping reduces damage but is insufficient; ATTR-AWARE is 80× more effective than the best structural defense.

Experiment	**BFS Damage**	**Attr Damage**	**Defense Rate**
CD 6K	68.9%	14.8%	78.5%
CD 32K	79.0%	15.8%	80.0%
CD 64K	78.5%	15.0%	80.9%
Cross-task (F1)	−18.2pp	+0.0pp	100.0%
Adv 6K / 1f	5.2%	0.0%	100.0%
Adv 6K / 5f	39.8%	0.3%	99.3%
Adv 32K / 1f	38.1%	0.0%	100.0%
Adv 32K / 5f	57.3%	1.0%	98.2%

Table 10: Defense effectiveness across all experiments.

6.3 Effectiveness Across All Experiments

7 Related Work

Memory systems and invalidation. As described in section 2, modern AI memory systems maintain entity graphs for retrieval [Chheda et al., 2025, Jiang et al., 2026, Xu et al., 2025, Rasmussen, 2025] and some detect factual conflicts at the individual-edge level [Chheda et al., 2025, Rasmussen, 2025]. A recent survey of graph-augmented agent memory [Yang et al., 2026] catalogs these architectures but does not address invalidation propagation. All graph-maintaining systems use graphs to *retrieve* relevant memories; we study what happens when graphs are used to *propagate invalidation*—a capability none currently implement, yet one that is architecturally inevitable as memory scales demand consistency beyond individual facts.

Knowledge editing and ripple effects. Knowledge editing modifies LLM parameters to update specific facts without full retraining. Recent work reveals that such edits produce non-local consequences: the RippleEdits benchmark [Cohen et al., 2024] finds that state-of-the-art editors achieve only 38–66% accuracy on logical consequences of edits, and Hase et al. [2024] argue that model editing fundamentally conflicts with belief revision desiderata. ChainEdit [Dong et al., 2025] addresses this by propagating edits through logical rule chains, achieving 30%+ improvement on downstream consistency. GradSim [Qin et al., 2024] predicts which facts will be affected by an edit via gradient similarity, revealing that ripple effects follow predictable but complex patterns. The ripple effect phenomenon is directly analogous to our collateral damage finding—both demonstrate that local updates produce non-local consequences—but the mechanisms differ fundamentally: knowledge editing operates on model parameters, whereas our work studies propagation in external, non-parametric memory stores where graph topology, not gradient geometry, determines the damage pattern.

Machine unlearning and graph unlearning. Machine unlearning removes the influence of specific training data from learned models, with methods ranging from exact retraining to approximate gradient-based approaches [Geng et al., 2025]. Graph unlearning extends this to graph-structured data, where node or edge removal must account for message-passing dependencies [Zhang et al., 2024]. A key finding from this literature is that removal of information from graphs can inadvertently degrade retained information—Ding et al. [2026] show that unlearning in GNNs amplifies memorization of remaining training examples, a structural parallel to our collateral damage result. Our work differs in target: graph unlearning modifies GNN model weights, whereas we study external memory stores traversed at inference time. The structural insight transfers—graph topology determines which information is collaterally affected—but the intervention points and defense mechanisms are distinct.

Memory-specific adversarial attacks. A growing body of work targets the persistent memory of LLM-based agents. MINJA [Ding et al., 2025] demonstrates query-only memory injection that achieves 98% success without privilege escalation, exploiting similarity-based retrieval to place adversarial content near target queries. AgentPoison [Xi et al., 2024] shows that poisoning less than 0.1% of a long-term memory corpus yields 80%+ attack success across autonomous agent benchmarks. ER-MIA [Piehl et al., 2026] extends injection attacks to black-box persistent memory via similarity exploitation, and Sunil et al. [2026] formalize attack–defense frameworks for memory poisoning. MITRE ATLAS designates persistent memory manipulation as technique AML.T0080 [MITRE Corporation, 2023], reflecting its emergence as a recognized threat category. These attacks inject malicious *content* into memory. Our hub exploitation attack is structurally distinct: the attacker does not inject harmful content but triggers the system's own legitimate forgetting mechanism to destroy valid memories, weaponizing *infrastructure* rather than *data*.

Temporal decay. Mnemosyne [Johnston et al., 2025] applies per-memory temporal decay via reverse sigmoid functions inspired by Ebbinghaus forgetting curves [Ebbinghaus, 1885], where each memory fades independently of graph connections. Our work is complementary: temporal decay determines *how fast* individual memories fade, while graph-based propagation determines *which* memories are affected by a factual update. The two mechanisms could be composed, with temporal decay governing passive forgetting and attribute-aware propagation governing active invalidation.

Network resilience. Scale-free networks exhibit robust-yet-fragile behavior: high tolerance to random node failures but extreme vulnerability to targeted hub removal [Albert et al., 2000, Barabási and Albert, 1999]. Percolation theory formalizes this by predicting that heterogeneous degree distributions lower the critical fraction for network fragmentation [Cohen et al., 2000, Molloy and Reed, 1995]. Recent work on compounding vulnerability [Rosas-Casals et al., 2026] shows that hub removal simultaneously degrades both percolation connectivity and cascade robustness, producing damage that exceeds the sum of individual effects. Transportation network resilience research [Chen et al., 2007] adds the directional dimension that motivates our defense: link failures propagate along specific routes, not uniformly through the network. Our results empirically confirm these network science predictions in AI memory systems—the co-occurrence graph exhibits scale-free topology ($\alpha = 2.73$–2.96), and hub-targeted attacks produce the catastrophic fragmentation that theory predicts.

8 Discussion

8.1 Implications for AI Memory Safety

Our results establish **memory graph exploitation** as a distinct attack surface in AI systems, separate from prompt injection and data poisoning. Key implications:

1. Any system that adopts graph-based invalidation propagation—a logical extension of existing graph-augmented memory systems—will be vulnerable to hub exploitation unless propagation is semantically constrained
2. The attack is low-cost: 5 injected facts destroy up to 57.3% of stored knowledge (at 32K scale); even a single fact causes 38.1% damage at scale
3. The vulnerability worsens with scale: more stored knowledge means larger blast radius
4. Attr-Aware propagation provides a practical defense without significant overhead

8.2 Why Hub Entities Are Fundamental

Hub entities reflect genuine reality—the United States really does connect to thousands of facts. The problem is that **co-occurrence $\neq$ causal dependency**. This asymmetry is structural, not incidental: in any corpus of factual statements, a small number of entities (nations, organizations, public figures) will appear across diverse topical contexts, creating high-degree nodes whose connections span unrelated domains. Removing or limiting these hubs is not viable—they are essential for retrieval. Attr-Aware propagation resolves this tension by distinguishing genuine dependency chains from coincidental co-occurrence, preserving the hub's retrieval utility while neutralizing its role as a damage amplifier.

Generalizability beyond this benchmark. One might ask whether the observed heavy-tailed topology ($\alpha \approx 2.7$–3.0) is an artifact of MemoryAgentBench's content distribution. We argue it is not, because the mechanism is linguistic, not dataset-specific. Entity mention frequencies in natural language follow Zipf's law [Zipf, 1949]: a small number of entities (nations, organizations, public figures) account for a disproportionate share of mentions across any sufficiently diverse corpus. When memories are constructed from such text, co-occurrence graphs inherit this skew, producing hub-and-spoke structures regardless of domain. This is consistent with findings in semantic network research, where free-association graphs [Steyvers and Tenenbaum, 2005] and knowledge graphs [Barabási and Albert, 1999] independently exhibit scale-free degree distributions. While replication on additional benchmarks remains important (Limitation 2), the Zipfian origin of hub formation suggests our results generalize to any memory system built from natural-language interactions.

8.3 Limitations

1. **Entity extraction**: Rule-based extraction treats "Vladimir Putin" and "Putin" as separate entities. Stronger entity resolution would improve both modes.

2. **Single benchmark**: Results from MemoryAgentBench only. Replication on other benchmarks would strengthen findings.

3. **Statistical variance**: Multi-run statistics (7 runs at 6K, 19 paired observations) show high within-attack variance (CV $\sim$150%) due to hub structure variability across contexts. Pooled Wilcoxon signed-rank test achieves strong significance ($p = 0.0022$) and bootstrap 95% BCa CI $[2.82, 15.67]$pp excludes zero. A mixed-effects robustness check (Appendix C, Table 14) confirms that attack strength significantly modulates the BFS–Attr-Aware difference (5-fact: $p = 0.020$) while properly accounting for run-level variation.

4. **White-box assumption**: Primary adversarial results assume white-box access. Black-box attacks fail entirely (0% hit rate), which simultaneously bounds the threat and characterizes its nature: hub exploitation is a *system-level* vulnerability requiring architectural knowledge of graph structure, fundamentally distinct from content-injection attacks (e.g., MINJA) that succeed from external positions. This makes the attack harder to mount but also harder to detect, as it operates through legitimate maintenance pathways.

5. **32K EM floor**: Baseline EM at 32K is 0–4%, limiting observable EM changes from adversarial attacks at that scale.

8.4 Future Work

1. **Relation-type graph**: Replace entity co-occurrence with typed-relation graphs (e.g., `president_of`, `located_in`) and compare propagation behavior. Typed edges may naturally suppress hub formation by requiring semantic specificity for connection.

2. **Learned propagation**: Replace rule-based Attr-Aware with a GNN-based propagation policy that learns which downstream facts to decay from training data.

3. **Adaptive depth**: Adjust propagation depth based on local graph topology (hub proximity, local density) rather than using a fixed global depth.

4. **Cross-system validation**: Apply the same attack methodology to production memory systems (Mem0^g, Zep) once they implement graph-based invalidation propagation.

9 Conclusion

As AI memory systems evolve from flat stores to graph-augmented architectures, graph-based invalidation propagation is the inevitable next step for maintaining fact consistency at scale. This work is, to our knowledge, the first to implement and systematically analyze graph-based invalidation propagation in external non-parametric memory stores, revealing that naive BFS propagation through entity co-occurrence graphs is **catastrophically destructive** (69–79% damage), **cross-task toxic** (F1 -10.1pp on unrelated retrieval), and **adversarially weaponizable** (5 fake facts $\rightarrow$ 57.3% memory destruction).

Attr-Aware Selective Propagation, inspired by directional link failure models in transportation engineering, provides consistent defense across all threat vectors (78–100% effectiveness) with negligible overhead. Graph topology analysis reveals the underlying mechanism: co-occurrence graphs exhibit heavy-tailed degree distributions consistent with scale-free properties ($\alpha \approx 2.7$–3.0), making them structurally robust to random failures but catastrophically vulnerable to targeted hub attacks—a prediction from percolation theory that our empirical results corroborate.

The key insight: **propagation must follow causal dependency chains, not structural co-occurrence**—a principle well-established in transportation network resilience but previously unapplied to AI memory systems.

A Experimental Configuration

Component	Specification
LLM	google/gemma-4-31B-it (max_tokens=10, temp=0.0)
Embedding	BAAI/bge-m3 (dim=1024)
Vector search	NumPy cosine similarity, RRF fusion ($k = 60$)
Graph	NetworkX entity co-occurrence (DiGraph)
Storage	SQLite in-memory
Benchmark	ai-hyz/MemoryAgentBench (ICLR 2026)
Context sizes	6K, 32K, 64K (FactConsolidation), 64K (EventQA)
Propagation	depth=2, decay_per_hop=0.5
Adversarial	1/3/5 fake facts × BFS/Attr × 6K/32K

Table 11: Experimental configuration.

B Hub Concentration Analysis

Scale	Top 1%	Top 5%	Top 10%	Top 20%
6K	7.3%	18.2%	27.3%	42.4%
32K	15.5%	29.8%	39.2%	53.0%
64K	19.8%	34.6%	43.8%	56.8%

Table 12: Hub entity concentration: percentage of total edges controlled by top-N% nodes. Inequality increases with scale.

C Multi-run Statistical Analysis

We run the 6K adversarial attack experiment 7 times (19 paired observations across attack strengths 1/3/5) to assess reproducibility.

Robustness check: mixed-effects model. To address the limitation of pooling across attack strengths, we fit a linear mixed-effects model with attack strength as a fixed effect and experimental run as a random intercept:

$$\Delta\text{damage}_{ij} = \beta_0 + \beta_1 \cdot \mathbb{1}[\text{3-fact}] + \beta_2 \cdot \mathbb{1}[\text{5-fact}] + u_j + \varepsilon_{ij}$$

where $u_j \sim \mathcal{N}(0, \sigma_u^2)$ captures run-level variation.

Attack	Mode	Damage (mean±std)	Cohen's d	p (Welch)	p (perm.)
1 fact	BFS Attr-Aware	1.6 ± 2.5 0.0 ± 0.1	0.95	0.126	**0.031**
3 facts	BFS Attr-Aware	8.9 ± 13.8 -0.1 ± 0.4	0.93	0.169	0.125
5 facts	BFS Attr-Aware	13.3 ± 20.6 -0.1 ± 0.7	0.92	0.171	0.125
Pooled (19 pairs)			0.66		
Wilcoxon signed-rank					$W = 86.0$, $p = 0.0022$
Bootstrap 95% BCa CI					$[2.82, 15.67]$ pp (excludes 0)

Table 13: Multi-run statistics (6K, 7 runs, 19 paired observations). Individual attack strengths show large effect sizes but limited sample sizes; pooled Wilcoxon signed-rank test achieves strong significance ($p = 0.0022$). Bootstrap 95% BCa CI excludes zero, confirming reliable BFS–Attr-Aware difference.

Fixed Effect	Coef.	SE	z	p
Intercept (1-fact baseline)	1.66	5.09	0.33	0.744
3-fact vs 1-fact	7.27	5.00	1.45	0.146
5-fact vs 1-fact	11.63	5.00	2.32	**0.020**
Random intercept $\hat{\sigma}_u^2$				104.17
N observations / groups				19 / 7

Table 14: Linear mixed-effects model: Δdamage $\sim C$(attack_strength) + (1|run). The 5-fact attack produces significantly greater BFS–Attr-Aware damage difference ($p = 0.020$). The large random intercept variance ($\hat{\sigma}_u^2 = 104.17$) reflects substantial between-run variability, explaining why individual attack strengths are underpowered when tested separately.

Negative damage values. ATTR-AWARE occasionally exhibits small negative damage values (-0.1 to -1.5%), which might appear paradoxical. Code inspection reveals two sources: (1) ATTR-AWARE's conflict resolution mechanism *replaces* outdated facts with attacker-injected facts via the `invalidate_and_replace` pathway, resetting the replacement memory's weight to 1.0. If the original fact had already decayed below 1.0, this replacement *decreases* the count of memories below the threshold, producing a negative Δdecayed. (2) A minor measurement artifact from natural time-based decay between pre- and post-attack snapshots (< 0.3pp). Both effects are bounded and confirm that ATTR-AWARE does not amplify attack damage—in the worst case, it inadvertently "refreshes" a small number of decayed memories.

3-fact condition significance. We note that the 3-fact attack condition does not reach statistical significance individually (permutation $p = 0.125$), consistent with its position between the barely-powered 1-fact ($p = 0.031$) and the significant 5-fact mixed-effects result ($p = 0.020$). With $n = 6$ paired observations, this condition is underpowered—a recognized limitation of our pilot-scale experimental design. The consistent direction of effect (BFS damage > 0 in 4/6 runs) and medium effect size (Cohen's $d = 0.93$) suggest the effect is real but would require additional runs to confirm at conventional significance levels.

D Graph Construction Details

This appendix provides full specification of the entity co-occurrence graph construction pipeline, enabling reproduction of our experimental results.

D.1 Entity Extraction

We employ a rule-based entity extractor optimized for FactConsolidation data (no NER model or LLM calls during memorization):

1. **Proper noun detection:** Regex `\b[A-Z][a-z]+(\s+[A-Z][a-z]+)*\b` captures capitalized word sequences. Stop-words (the, a, an, is, was, are, were, has, have) are excluded.
2. **"The X of Y" pattern:** Regex `the\s+(\w+)\s+of\s+([A-Z]\w+(\s+[A-Z]\w+)*)` extracts the object entity Y from relational sentences.

Example: *"The capital of France is Paris."* $\rightarrow$ entities {France, Paris}.

D.2 Co-occurrence Edge Construction

For each memorized fact containing extracted entities $\{e_1, e_2, \ldots, e_n\}$, we add a `co_occurs` edge between every pair (e_i, e_j) for $i < j$. This creates a clique of size $\binom{n}{2}$ per fact. All edges are stored in a NetworkX `DiGraph` but treated as undirected for BFS propagation (both in- and out-neighbors are traversed).

D.3 Subject-Key Generation for Conflict Detection

Each fact is assigned a deterministic *subject-key* for conflict detection:

1. **Primary pattern:** `the\s+(\w+)\s+of\s+(...)\s+(is|was|are|were)` $\rightarrow$ key = `entity:attribute`. Example: *"The president of Russia is Putin"* $\rightarrow$ `Russia:president`.
2. **Secondary patterns:** `is the X of Y` $\rightarrow$ `Y:X`; relation-specific patterns (born in, lives in, plays for, etc.) $\rightarrow$ `subject:relation`.
3. **Fallback:** First entity + structure hash (after replacing all entity names with placeholders).

When a new fact maps to the same subject-key as an existing fact, the system detects a *factual conflict*: the old memory is invalidated and graph propagation is triggered from the shared entity node.

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